Lecture plan · for the lecturer

Teaching machine learning at university: unit check

AI · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Optimisation
- Backpropagation
- Attention
- Evaluation and assessment

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change assessment?

Exit ticket

- One change to my ML course this term, and the evidence I will collect.

Try it in your next lecture
Bring one revised ML assignment and its rubric to your next department meeting.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **Loss θ^2, $\theta = 5$, $\eta = 0.1$. After one step, θ =**

◯ 4 ◯ 4.9 ◯ 3 ◯ 0

2 **The negative gradient points directly at the minimum.**

◯ True ◯ False

3 **Apply the chain rule.**

$\partial L/\partial a = 4$ and $\partial a/\partial w = 0.5$, so $\partial L/\partial w =$ _____ .

Word bank: 2 · 4.5 · 8

4 **Which reduce vanishing gradients? Choose all that apply.**

☐ ReLU ☐ Residual connections ☐ Deeper sigmoid stacks

5 **d = 256. Raw attention scores are divided by**

◯ 16 ◯ 256 ◯ 8